

CS201: Data Structures II Project Proposal

Project Title: Epidemic Outbreak Detection System

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1 Abstract

Provide a brief summary of your project, including its purpose and significance.

This project aims to develop an Epidemic Outbreak Detection System that analyzes real-world disease datasets to identify patterns and predict potential outbreak regions. The system utilizes a Sparse Matrix to efficiently store large-scale epidemiological data and a Decision Tree to classify regions into categories such as Safe, Warning, and Outbreak. A priority queue (heap) is used to efficiently rank regions based on risk levels, allowing quick identification of high-risk areas. A KD-tree is also used to organize multi-dimensional features such as growth rate and population, enabling efficient comparison of regions. The results are displayed through a terminal-based interface and visualized using graphs generated via SFML, with GNUplot as a backup option.

2 Introduction

Introduce the problem statement and explain why this topic is relevant to data structures.

Large health datasets are often difficult to manage because they contain many zero values, which wastes memory. To solve this, we use a Sparse Matrix to store only useful data. The system then analyzes trends such as growth rate and recent increases in cases. A Decision Tree is used to classify regions as Safe, Warning, or Outbreak, while a priority queue helps quickly identify high-risk regions. Additionally, a KD-tree is used to organize regions based on multiple features, enabling efficient comparison between regions. Overall, the project shows how data structures can be used to efficiently store and analyze real-world epidemic data.

3 Objectives

List the main goals of your project. These should be clear and concise.

1. To store epidemic data efficiently using a Sparse Matrix by keeping only non-zero.
2. To load and process real-world dataset (region, day, cases, population)
3. We will analyze trends such as growth rate and recent increases in cases
4. To detect short-term trends (e.g., cases increasing continuously over last few days)
5. Classify regions as Safe, Warning, or Outbreak using a Decision Tree
6. To identify high-risk regions quickly using a priority queue (heap)
7. To visualize case trends using graphs.
8. To organize regions using a KD-tree for efficient multi-feature comparison (e.g., growth rate and population).

4 Methodology

Describe the approach you will take to implement the project. Include:

- The data structures you plan to use
 1. Sparse Matrix (to store only non-zero data)
 2. Vector/Array (to store cases over time)
 3. Priority Queue/Heap (to find high-risk regions)
 4. KD-tree (to compare regions using growth and population)
- Algorithms you will implement
 1. Growth rate calculation
 2. Decision Tree for classification (Safe, Warning, Outbreak)
 3. Sliding window for trend detection (last few days increase)
- Tools and technologies required
 1. C++
 2. SFML for visualization
 3. GNUplot as backup for graphs
- Expected challenges
 1. Handling real-world dataset (messy data)
 2. Implementing Sparse Matrix correctly
 3. Setting correct thresholds for classification
 4. Implementing KD-tree for multi-feature comparison

5 Expected Outcomes

Explain what results you expect to achieve from the project.

1. A system that can store epidemic data efficiently
2. Ability to classify regions as Safe, Warning, or Outbreak
3. Detection of short-term trends (increasing cases)
4. Identification of high-risk regions quickly ($O(1)$ retrieval and $O(\log n)$ insertion)
5. Comparison of regions using multiple features
6. Graphs to visualize case trends clearly

6 References

List any books, papers, or online resources you will use in your project.

1. Dinesh Mehta, Handbook of Data Structures and Applications
2. Kaggle Datasets
3. GitHub
4. GeeksforGeeks